

Week -2 Module

① Fraud Transaction Detection (Linear Search)

Problem :- A bank stores transaction IDs in the order they occur. Since the transactions are not stored, the security team needs to check if a suspicious transaction ID exists.

Input Format

→ Solo Linear Search :- It is a simplest searching technique. It checks each element one by one from the beginning until the target element is found or the list ends.

In
Java

```
import java.util.Scanner;
```

```
public class Main {
```

```
    public static void main (String[] args) {
```

```
        Scanner sc = new Scanner (System.in);
```

```
        int n = sc.nextInt();
```

```
        int[] transactions = new int[n];
```

```
        for (int i = 0; i < n; i++) {
```

```
            transactions[i] = sc.nextInt();
```

```
        }
```

```
        int suspiciousId = sc.nextInt();
```

```
        boolean found = false;
```

```
        for (int i = 0; i < n; i++) {
```

```
            if (transactions[i] == suspiciousId) {
```

```
                found = true;
```

```
                break;
```

```
            }
```

```
        }
```

```
        if (found) {
```

```
            System.out.println ("Fraud Transaction Found")
```

```
        } else {
```

```
            System.out.println ("Fraud Transaction Not Found")
```

```
        }
```

```
        sc.close();
```

```
File Edit Selection View Tools Window Help
Main.java x
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] transactions = new int[n];
9
10        for (int i = 0; i < n; i++) {
11            transactions[i] = sc.nextInt();
12        }
13
14        int suspiciousId = sc.nextInt();
15
16        boolean found = false;
17    }
18 }
19
20 PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
21
22 Syntax error, insert "AssignmentOperator Expression" to complete Assignment
23 Syntax error, insert ";" to complete Statement
24 The left-hand side of an assignment must be a variable
25
26 at Main.main(Main.java:11)
27 PS C:\C++> ^C
28 PS C:\C++>
29 PS C:\C++> C: cd 'C:\C++'; & 'C:\Program Files\Java\jre1.8.0_461\bin\java.exe' '-cp' 'C:\Users\golue\AppData\Roaming\Code\User\workspaceStorage\ae37b18392cc563349521a1df282a430\redhat
30 .java\jdk_ms\c++_759c632b\bin' 'Main'
31 5
32 1201
33 1305
34 1450
35 1408
36 1502
37 1450
38 Fraud Transaction Found
39 PS C:\C++>
40
41 (Windows) Launch (C++) Java: Ready
42 Ln 11, Col 44 Spaces: 4 UTF-8 CRLF {} Java Go Live
43 27°C Mostly sunny 18:15 21-05-2025
```


2.) Server Log Search (Binary Search)

A cloud company stores server log IDs in sorted order. Engineers need to quickly verify if a log ID exists.

Write a program using Binary Search

Solⁿ Binary Search:- It is an efficient algorithm to find an element in a sorted array. It works by repeatedly dividing the search interval in half.

In Java

```
import java.util.Scanner;
```

```
public static void main {
```

```
    public static void main (String [] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int n = nextInt sc.nextInt();
```

```
        int [] logs = new int[n];
```

```
        for (int i = 0; i < n; i++) {
```

```
            int log[i] = sc.nextInt();
```

```
        }
```

```
        int target = sc.nextInt();
```

```
        int left = 0;
```

```
        int right = n - 1;
```

```
        boolean found = false;
```

```
        while (left <= right) {
```

```
            int mid = left + (right - left) / 2;
```

```
            if (logs[mid] == target) {
```

```
                found = true;
```

```
                break;
```

```
            } else if (logs[mid] < target) {
```

```
                left = mid + 1;
```

```
            } else {
```

```
                right = mid - 1;
```

```
            }
```

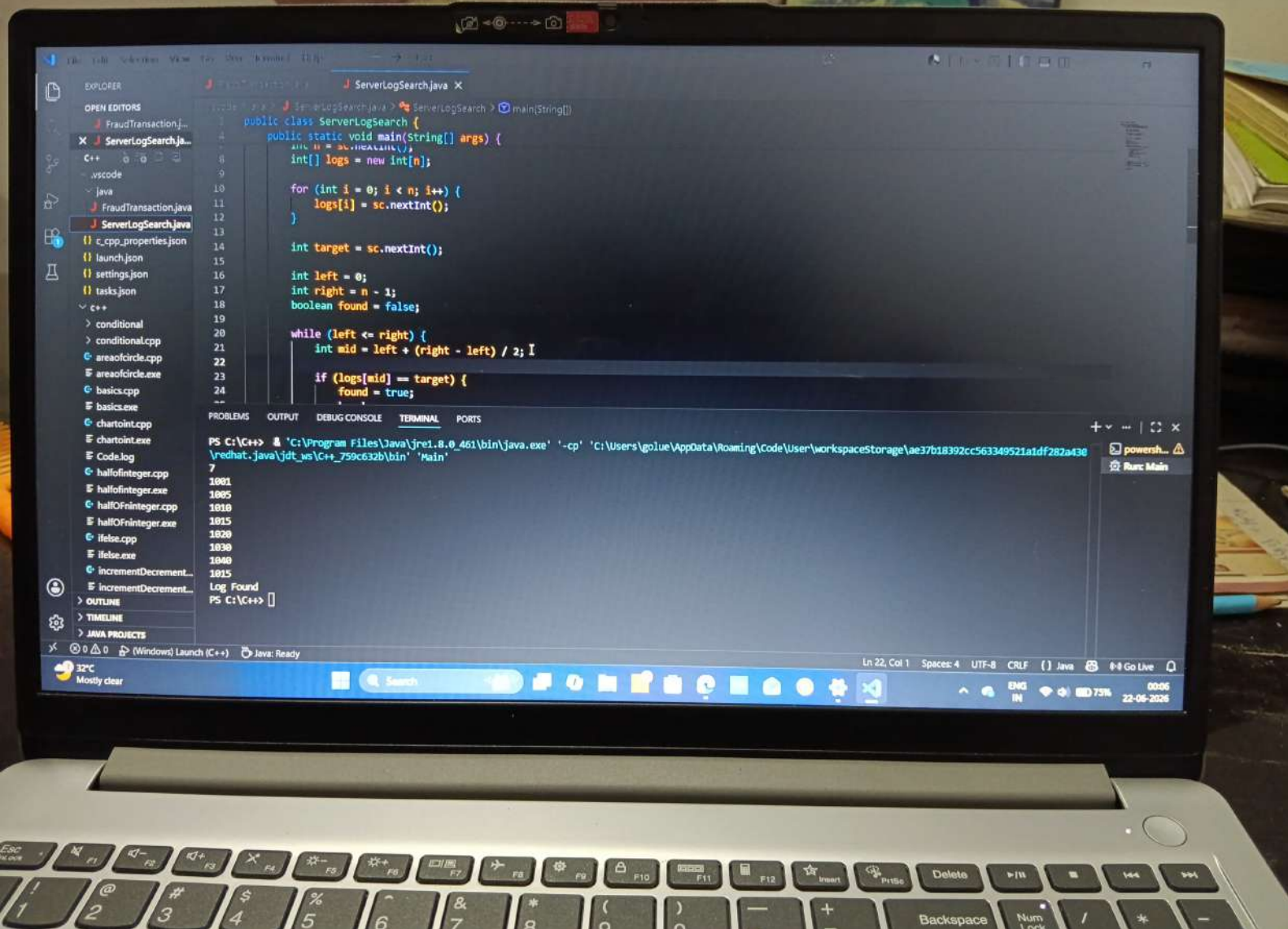
```
        } if (found) {
```

```
            System.out.println("Log found");
```

```
        } else {
```

```
            System.out.println("Log not found");
```

```
        } sc.close();
```



3.) Sort Employee Salaries (Bubble Sort)

A company wants to sort employee salaries in ascending order for payroll analysis.

Write a program using Bubble Sort.

Solⁿ It is a simple sorting algorithm that repeatedly compares adjacent elements and swaps them if they are in the wrong order.

In Java

```
import java.util.Scanner;
```

```
public static void main {
```

```
    public static void main (String [] args) {
```

```
        Scanner sc = new Scanner (System.in);
```

```
        int n = sc.nextInt();
```

```
        int [] salaries = new int[n];
```

```
        for (int i = 0; i < n; i++) {
```

```
            salaries[i] = sc.nextInt();
```

```
        }
```

```
        for (int i = 0; i < n - 1; i++) {
```

```
            for (int j = 0; j < n - 1 - i; j++) {
```

```
                if (salaries[j] > salaries[j + 1]) {
```

```
                    int temp = salaries[j];
```

```
                    salaries[j] = salaries[j + 1];
```

```
                    salaries[j + 1] = temp;
```

```
                }
```

```
            }
```

```
        }
```

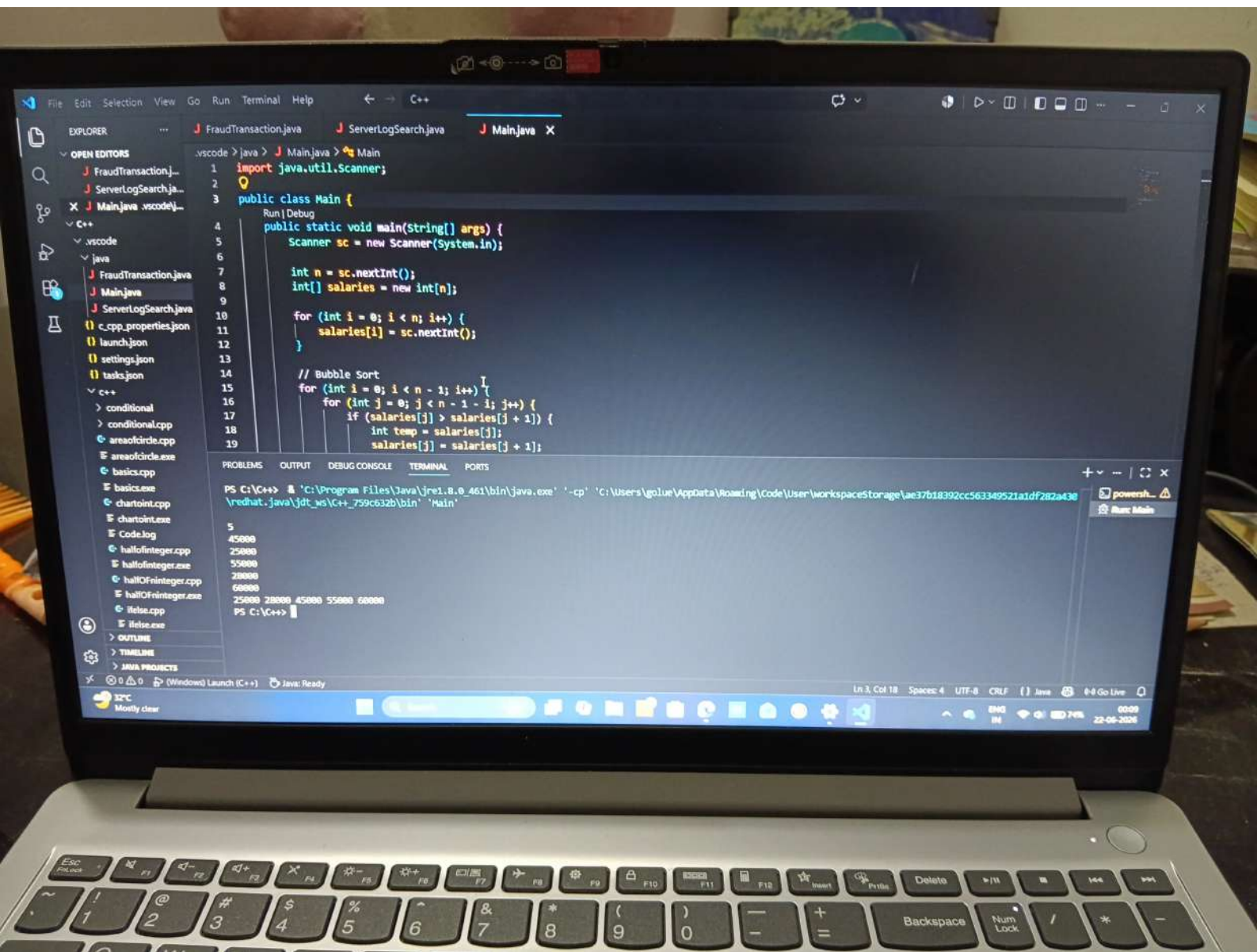
```
        System.out.println(salaries[i] + " ");
```

```
    }
```

```
    sc.close();
```

```
}
```

```
}
```



```
.vscode > java > J Main.java > Main
1 import java.util.Scanner;
2
3 public class Main {
4     Run | Debug
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7
8         int n = sc.nextInt();
9         int[] salaries = new int[n];
10
11         for (int i = 0; i < n; i++) {
12             salaries[i] = sc.nextInt();
13         }
14
15         // Bubble Sort
16         for (int i = 0; i < n - 1; i++) {
17             for (int j = 0; j < n - 1 - i; j++) {
18                 if (salaries[j] > salaries[j + 1]) {
19                     int temp = salaries[j];
20                     salaries[j] = salaries[j + 1];
21                     salaries[j + 1] = temp;
22                 }
23             }
24         }
25     }
26 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\C++ > 'C:\Program Files\Java\jre1.8.0_461\bin\java.exe' '-cp' 'C:\Users\goLue\AppData\Roaming\code\User\workspaceStorage\ae37b18392cc563349521a1df282a436\redhat.java\jdk_15\c++_759c532b\bin' 'Main'
5
45000
25000
55000
28000
60000
25000 28000 45000 55000 60000
PS C:\C++ >
```

32°C Mostly clear (Windows) Launch (C++) Java: Ready

Ln 3, Col 18 Spaces: 4 UTF-8 CRLF Java 6-4 Go Live

0000

22-05-2026

4.) Sort Website Response Time (Insertion Sort)

A web monitoring system records website response times. New data comes continuously and must be inserted into the correct position.

Solⁿ Insertion Sort :- It is a sorting algorithm that builds the sorted array one element at a time by inserting each element into its correct position.

In Java

```
import java.util.Scanner;
```

```
public class Main {
```

```
    public static void main (String[] args) {
```

```
        Scanner sc = new Scanner (System.in);
```

```
        int n = sc.nextInt();
```

```
        int[] responseTimes = new int[n];
```

```
        for (int i = 0; i < n; i++) {
```

```
            responseTimes[i] = sc.nextInt();
```

```
        }
```

```
        for (int i = 1; i < n; i++) {
```

```
            int key = responseTimes[i];
```

```
            int j = i - 1;
```

```
            while (j >= 0 && responseTimes[j] > key) {
```

```
                responseTimes[j+1] = responseTimes[j];
```

```
                j--;
```

```
            }
```

```
            responseTimes[j+1] = key;
```

```
        }
```

```
        for (int i = 0; i < n; i++) {
```

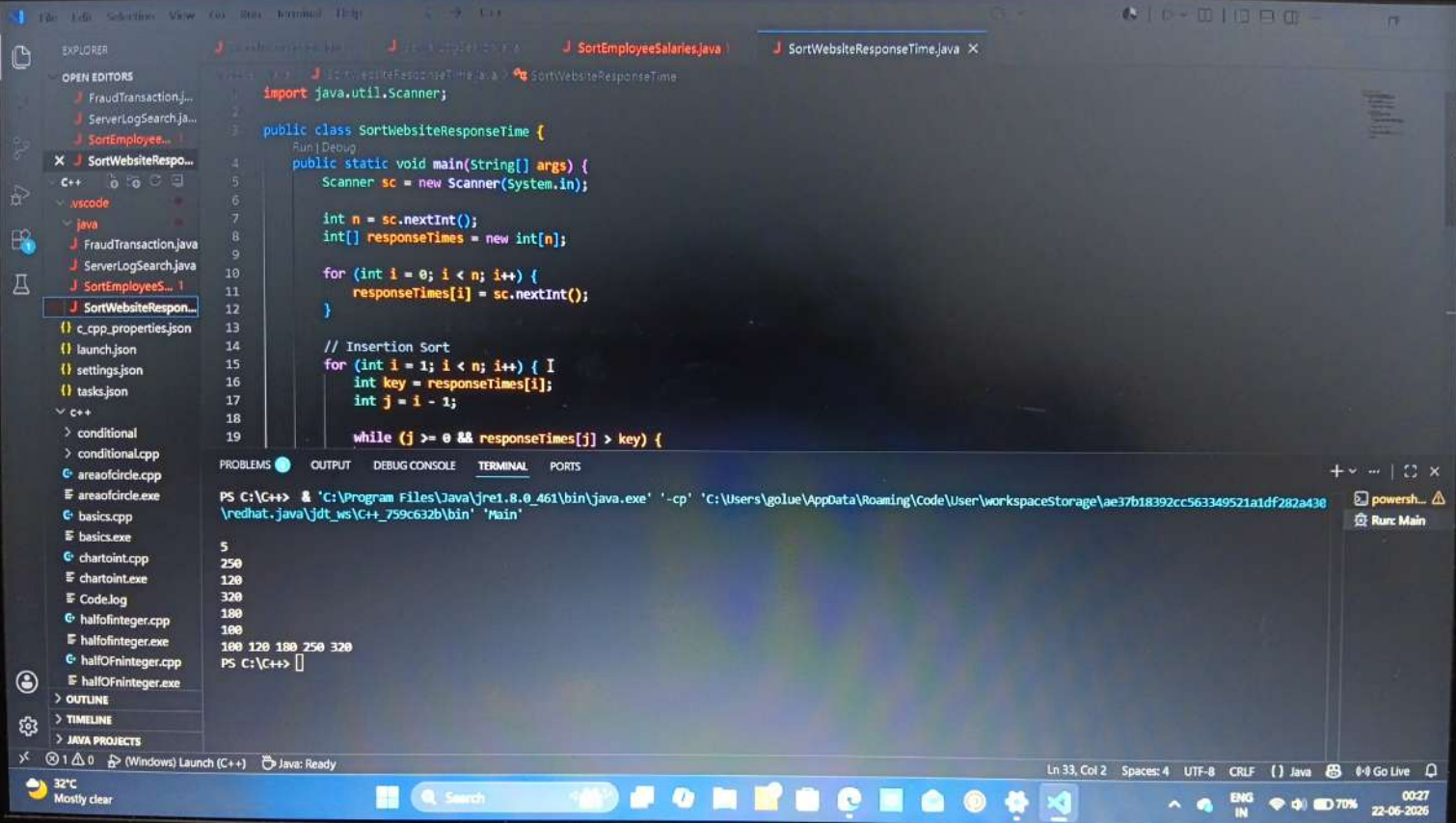
```
            System.out.print (responseTimes[i] + " ");
```

```
        }
```

```
        sc.close();
```

```
    }
```

```
}
```



5) Prioritize Bug Reports (Selection Sort)
 A Software Company assigns priority scores to bug reports. To resolve bugs efficiently, they want to sort priorities in ascending order.

Solⁿ - Selection Sort :- It is a sorting algorithm that

repeatedly finds the smallest element from the unsorted part of the array and place it at the beginning.

In Java

```
import java.util.Scanner;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();
```

```
        int[] priorities = new int[n];
```

```
        for (int i = 0; i < n; i++) {
```

```
            priorities[i] = sc.nextInt();
```

```
        }
```

```
        for (int i = 0; i < n-1; i++) {
```

```
            int minIndex = i;
```

```
            for (int j = i+1; j < n; j++) {
```

```
                if (priorities[j] < priorities[minIndex]) {
```

```
                    minIndex = j;
```

```
                }
```

```
            }
```

```
            int temp = priorities[i];
```

```
            priorities[i] = priorities[minIndex];
```

```
            priorities[minIndex] = temp;
```

```
        }
```

```
        for (int i = 0; i < n; i++) {
```

```
            System.out.print(priorities[i] + " ");
```

```
        }
```

```
    }
```

```
    Scanner sc;
```

```
}
```

